Advanced Network Security and Architectures

METI

Report Module 2 [EN]

Authors:
José Oliveira (103771) jose.novais.oliveira@tecnico.ulisboa.pt
Afonso Mendes (116024) afonsofmendes@tecnico.ulisboa.pt
Tiago Videira (116069) tiago.g.videira@tecnico.ulisboa.pt

Group 3

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2.1 Introduction

In this experiment, IPSec tunnels using IKEv1 with Pre-Shared Keys (PSK) were analyzed. Both AH and ESP protocols were tested in order to observe the establishment of IPSec Security Associations (SAs), OSPF traffic protection, and the exchanged ISAKMP messages during tunnel negotiation.

2.2 AH Tunnel Operation

Initially, the IPSec tunnel was configured using the Authentication Header (AH) protocol. ICMP traffic exchanged between PC1 and PC2 was captured using Wireshark.

Figure 1 shows an AH-protected ICMP packet. The AH header includes fields such as the Security Parameters Index (SPI), Sequence Number, and Integrity Check Value (ICV). The SPI identifies the Security Association being used, the sequence number prevents replay attacks, and the ICV guarantees packet integrity and authentication.

Unlike ESP, AH does not encrypt the payload. Therefore, the original ICMP packet remains visible after the AH header.

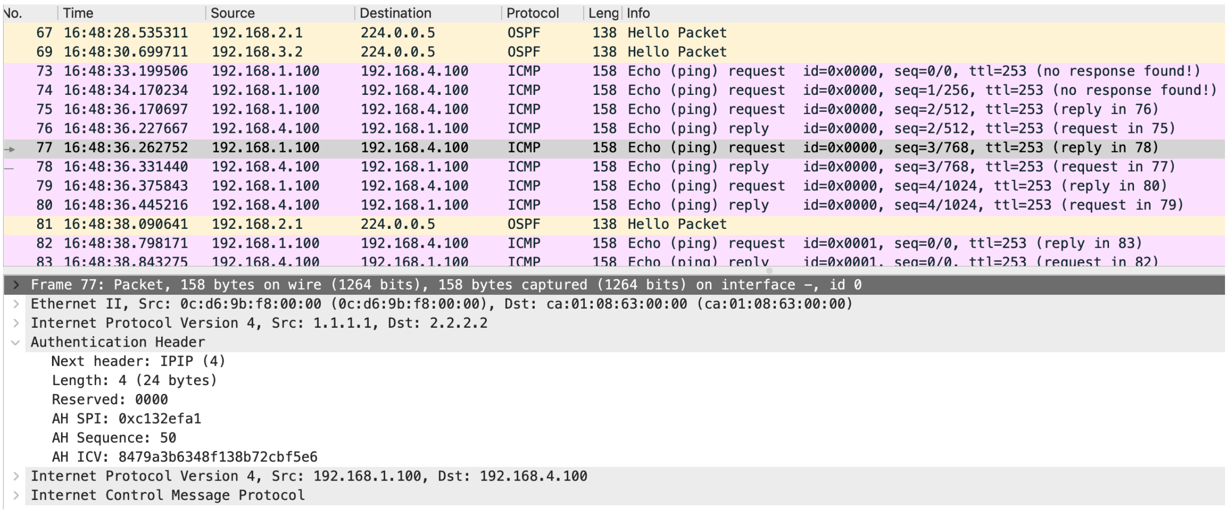


Figure 1: AH protecting ICMP traffic between PC1 and PC2

2.3 OSPF Traffic Analysis

OSPF traffic exchanged through the tunnel was also analyzed.

Figure 2 shows a normal OSPF Hello packet before IPSec protection. The packet includes the Router ID, Area ID, Hello Interval, Dead Interval, and neighbor information required for adjacency establishment.

```
161 16:50:27.086521   200.1.1.10         224.0.0.5          OSPF        94 Hello Packet
162 16:50:27.346867   200.1.1.1          224.0.0.5          OSPF        94 Hello Packet
163 16:50:28.006268   192.168.3.2        224.0.0.5          OSPF       138 Hello Packet
166 16:50:31.079919   192.168.2.1        224.0.0.5          OSPF       138 Hello Packet
167 16:50:36.734564   200.1.1.10         224.0.0.5          OSPF        94 Hello Packet

> Frame 161: Packet, 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface -, id 0
> Ethernet II, Src: ca:01:08:63:00:00 (ca:01:08:63:00:00), Dst: IPv4mcast_05 (01:00:5e:00:00:05)
> Internet Protocol Version 4, Src: 200.1.1.10, Dst: 224.0.0.5
v Open Shortest Path First
  v OSPF Header
      Version: 2
      Message Type: Hello Packet (1)
      Packet Length: 48
      Source OSPF Router: 200.2.2.10
      Area ID: 0.0.0.0 (Backbone)
      Checksum: 0xc77c [correct]
      Instance ID: Base IPv4 Unicast Instance (0)
      Auth Type: Null (0)
      Auth Data (none): 0000000000000000
  v OSPF Hello Packet
      Network Mask: 255.255.255.0
      Hello Interval [sec]: 10
    > Options: 0x12, (L) LLS Data block, (E) External Routing
      Router Priority: 1
      Router Dead Interval [sec]: 40
      Designated Router: 200.1.1.10
      Backup Designated Router: 200.1.1.1
      Active Neighbor: 200.1.1.1
  > OSPF LLS Data Block
```

Figure 2: OSPF Hello packet without IPSec protection

Figure 3 shows the same OSPF traffic protected using AH. The Authentication Header appears before the OSPF packet, proving that the packet integrity and authentication are protected. However, the OSPF payload itself remains visible because AH does not provide confidentiality.

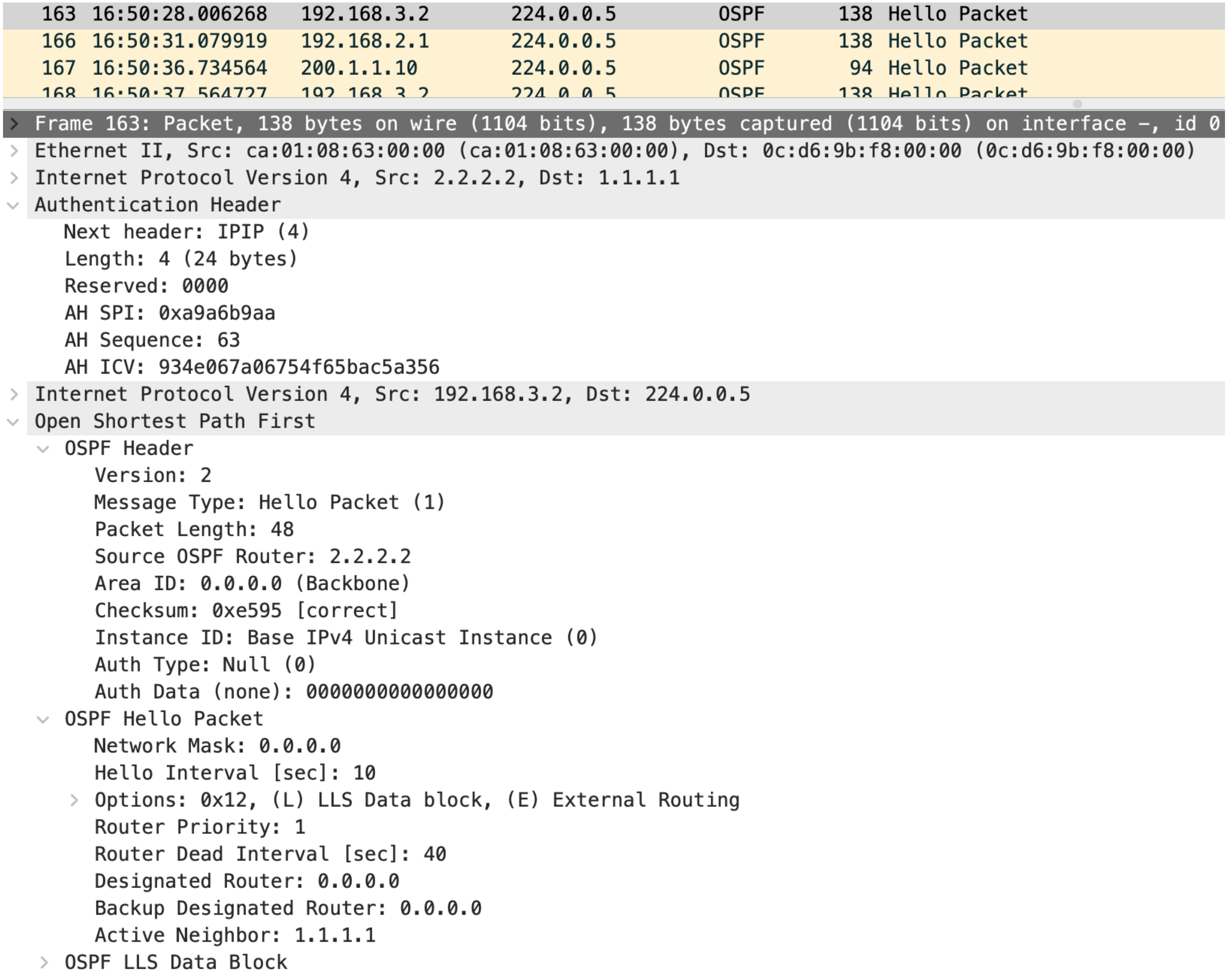


Figure 3: OSPF Hello packet protected with AH

2.4 IKEv1 and ISAKMP Negotiation

The IPSec Security Associations were cleared using the command:

```
clear crypto session
```

This operation forced the routers to delete the existing IPSec tunnel and establish a new one. During this process, Wireshark captured the exchanged ISAKMP packets.

2.4.1 ISAKMP Informational Messages

Before re-establishing the tunnel, ISAKMP Informational messages were exchanged between peers. These messages are used to maintain or delete Security Associations.

Figure 4 shows an Informational exchange containing encrypted data.

```
41 17:24:43.971851   1.1.1.1          2.2.2.2          ISAKMP    118 Informational
42 17:24:44.038276   1.1.1.1          2.2.2.2          ISAKMP    118 Informational
43 17:24:44.264407   1.1.1.1          2.2.2.2          ISAKMP    134 Informational
44 17:24:45.115767   2.2.2.2          1.1.1.1          ISAKMP    134 Informational
45 17:24:45.397728   1.1.1.1          2.2.2.2          ISAKMP    210 Identity Protection (Main M
> Frame 41: Packet, 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface -, id 0
> Ethernet II, Src: 0c:d6:9b:f8:00:00 (0c:d6:9b:f8:00:00), Dst: ca:01:08:63:00:00 (ca:01:08:63:00:00)
> Internet Protocol Version 4, Src: 1.1.1.1, Dst: 2.2.2.2
> User Datagram Protocol, Src Port: 500, Dst Port: 500
v Internet Security Association and Key Management Protocol
    Initiator SPI: 77ece4687be4e5cf
    Responder SPI: c2f86656ee8a5d4c
    Next payload: Hash (8)
  > Version: 1.0
    Exchange type: Informational (5)
  > Flags: 0x01
    Message ID: 0x48606521
    Length: 76
    Encrypted Data (48 bytes)
```

Figure 4: ISAKMP Informational message

2.4.2 IKE Phase 1 – Main Mode

IKE Phase 1 establishes the ISAKMP Security Association used to securely negotiate IPSec parameters.

Figure 5 shows the initial Main Mode exchange. The Security Association payload contains the proposed cryptographic algorithms and authentication methods supported by the initiator.

```
45 17:24:45.397728   1.1.1.1          2.2.2.2          ISAKMP    210 Identity Protection (Main Mode)
46 17:24:45.753871   2.2.2.2          1.1.1.1          ISAKMP    150 Identity Protection (Main Mode)
47 17:24:45.833812   1.1.1.1          2.2.2.2          ISAKMP    390 Identity Protection (Main Mode)
> Frame 45: Packet, 210 bytes on wire (1680 bits), 210 bytes captured (1680 bits) on interface -, id 0
> Ethernet II, Src: 0c:d6:9b:f8:00:00 (0c:d6:9b:f8:00:00), Dst: ca:01:08:63:00:00 (ca:01:08:63:00:00)
> Internet Protocol Version 4, Src: 1.1.1.1, Dst: 2.2.2.2
> User Datagram Protocol, Src Port: 500, Dst Port: 500
v Internet Security Association and Key Management Protocol
    Initiator SPI: 77ece468cdcf914d
    Responder SPI: 0000000000000000
    Next payload: Security Association (1)
  > Version: 1.0
    Exchange type: Identity Protection (Main Mode) (2)
  > Flags: 0x00
    Message ID: 0x00000000
    Length: 168
  > Payload: Security Association (1)
  > Payload: Vendor ID (13) : RFC 3947 Negotiation of NAT-Traversal in the IKE
  > Payload: Vendor ID (13) : draft-ietf-ipsec-nat-t-ike-07
  > Payload: Vendor ID (13) : draft-ietf-ipsec-nat-t-ike-03
  > Payload: Vendor ID (13) : draft-ietf-ipsec-nat-t-ike-02\n
```

Figure 5: IKEv1 Main Mode Security Association negotiation

Figure 6 shows the Diffie-Hellman exchange phase. The Key Exchange payload carries the public Diffie-Hellman values required to generate the shared secret key between both peers. The Nonce payload adds randomness to the generated keys, increasing security.

The Key Exchange payload is therefore responsible for securely generating a common secret without transmitting the secret itself over the network.

```
47  17:24:45.833812   1.1.1.1    2.2.2.2    ISAKMP   390  Identity Protection (Main Mode)
48  17:24:46.001470   2.2.2.2    1.1.1.1    ISAKMP   410  Identity Protection (Main Mode)
49  17:24:46.265562   1.1.1.1    2.2.2.2    ISAKMP   118  Identity Protection (Main Mode)

> Frame 47: Packet, 390 bytes on wire (3120 bits), 390 bytes captured (3120 bits) on interface -, id 0
> Ethernet II, Src: 0c:d6:9b:f8:00:00 (0c:d6:9b:f8:00:00), Dst: ca:01:08:63:00:00 (ca:01:08:63:00:00)
> Internet Protocol Version 4, Src: 1.1.1.1, Dst: 2.2.2.2
> User Datagram Protocol, Src Port: 500, Dst Port: 500
v Internet Security Association and Key Management Protocol
    Initiator SPI: 77ece468cdcf914d
    Responder SPI: c2f86656590ef065
    Next payload: Key Exchange (4)
  > Version: 1.0
    Exchange type: Identity Protection (Main Mode) (2)
  > Flags: 0x00
    Message ID: 0x00000000
    Length: 348
  > Payload: Key Exchange (4)
  > Payload: Nonce (10)
  > Payload: Vendor ID (13) : RFC 3706 DPD (Dead Peer Detection)
  > Payload: Vendor ID (13) : Unknown Vendor ID
  > Payload: Vendor ID (13) : XAUTH
  > Payload: NAT-D (RFC 3947) (20)
  > Payload: NAT-D (RFC 3947) (20)
```

Figure 6: Diffie-Hellman Key Exchange during IKE Phase 1

Figure 7 shows the final Main Mode exchanges where identity information and authentication data are transmitted in encrypted form. At this stage, the ISAKMP SA has already been established.

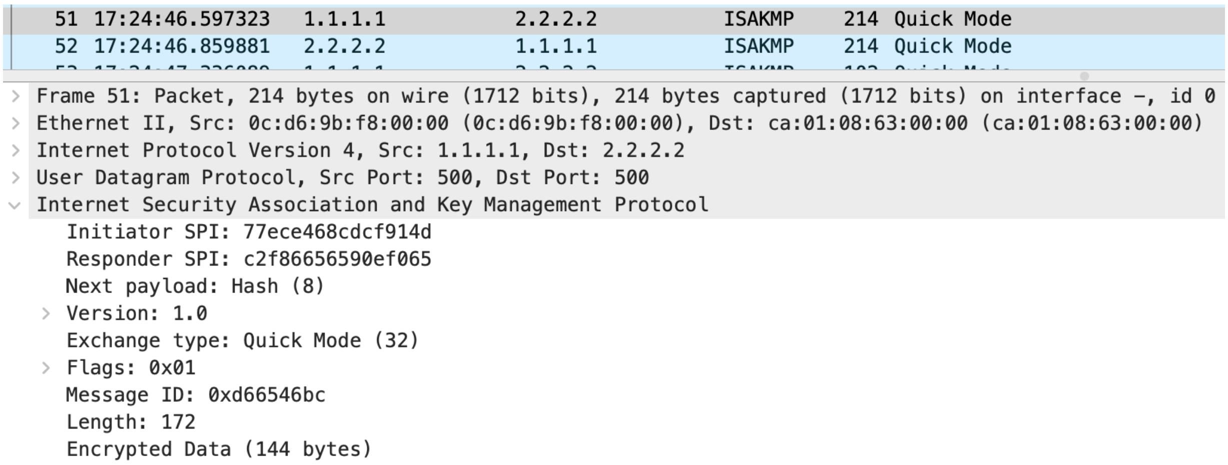
```
49  17:24:46.265562   1.1.1.1    2.2.2.2    ISAKMP   118  Identity Protection (Main Mode)
50  17:24:46.446131   2.2.2.2    1.1.1.1    ISAKMP   118  Identity Protection (Main Mode)
51  17:24:46.597323   1.1.1.1    2.2.2.2    ISAKMP   214  Quick Mode
52  17:24:46.859881   2.2.2.2    1.1.1.1    ISAKMP   214  Quick Mode

> Frame 49: Packet, 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface -, id 0
> Ethernet II, Src: 0c:d6:9b:f8:00:00 (0c:d6:9b:f8:00:00), Dst: ca:01:08:63:00:00 (ca:01:08:63:00:00)
> Internet Protocol Version 4, Src: 1.1.1.1, Dst: 2.2.2.2
> User Datagram Protocol, Src Port: 500, Dst Port: 500
v Internet Security Association and Key Management Protocol
    Initiator SPI: 77ece468cdcf914d
    Responder SPI: c2f86656590ef065
    Next payload: Identification (5)
  > Version: 1.0
    Exchange type: Identity Protection (Main Mode) (2)
  > Flags: 0x01
    Message ID: 0x00000000
    Length: 76
    Encrypted Data (48 bytes)
```

Figure 7: Identity protection during IKE Phase 1

2.4.3 IKE Phase 2 – Quick Mode

IKE Phase 2 negotiates the IPSec Security Associations used to protect data traffic.

Figure 8 shows the Quick Mode exchange responsible for negotiating IPSec parameters such as AH/ESP protocols, encryption algorithms, integrity algorithms, and Security Associations.



```
51 17:24:46.597323   1.1.1.1    2.2.2.2    ISAKMP   214 Quick Mode
52 17:24:46.859881   2.2.2.2    1.1.1.1    ISAKMP   214 Quick Mode
> Frame 51: Packet, 214 bytes on wire (1712 bits), 214 bytes captured (1712 bits) on interface -, id 0
> Ethernet II, Src: 0c:d6:9b:f8:00:00 (0c:d6:9b:f8:00:00), Dst: ca:01:08:63:00:00 (ca:01:08:63:00:00)
> Internet Protocol Version 4, Src: 1.1.1.1, Dst: 2.2.2.2
> User Datagram Protocol, Src Port: 500, Dst Port: 500
v Internet Security Association and Key Management Protocol
    Initiator SPI: 77ece468cdcf914d
    Responder SPI: c2f86656590ef065
    Next payload: Hash (8)
  > Version: 1.0
    Exchange type: Quick Mode (32)
  > Flags: 0x01
    Message ID: 0xd66546bc
    Length: 172
    Encrypted Data (144 bytes)
```

Figure 8: IKE Phase 2 Quick Mode negotiation

The router outputs confirmed the successful establishment of the tunnel. The ISAKMP state reached `QM_IDLE`, indicating that Phase 1 and Phase 2 negotiations completed successfully.

```
Router#show crypto isakmp sa

dst            src            state          conn-id status
1.1.1.1        2.2.2.2        QM_IDLE           1002 ACTIVE
2.2.2.2        1.1.1.1        QM_IDLE           1003 ACTIVE
```

The IPSec Security Associations were also verified using:

```
show crypto ipsec sa
```

The output confirmed active inbound and outbound AH Security Associations.

2.5 Negotiation Failure Analysis

Different cryptographic parameters were intentionally configured between peers in order to analyze negotiation failures.

Figure 9 shows repeated Main Mode and Quick Mode negotiations. Since the peers could not agree on compatible IPSec parameters, the Security Associations were not successfully established.

As a result, ICMP communication failed and no encrypted traffic was exchanged through the tunnel.

151	12:09:26.242043	1.1.1.1	2.2.2.2	ISAKMP	118	Informational
152	12:09:26.288903	1.1.1.1	2.2.2.2	ISAKMP	118	Informational
153	12:09:26.493217	1.1.1.1	2.2.2.2	ISAKMP	134	Informational
154	12:09:27.316934	1.1.1.1	2.2.2.2	ISAKMP	210	Identity Protection (Main Mode)
155	12:09:27.443088	2.2.2.2	1.1.1.1	ISAKMP	150	Identity Protection (Main Mode)
156	12:09:27.652128	1.1.1.1	2.2.2.2	ISAKMP	390	Identity Protection (Main Mode)
157	12:09:27.886886	2.2.2.2	1.1.1.1	ISAKMP	410	Identity Protection (Main Mode)
158	12:09:28.100810	1.1.1.1	2.2.2.2	ISAKMP	150	Identity Protection (Main Mode)
160	12:09:28.255179	2.2.2.2	1.1.1.1	ISAKMP	118	Identity Protection (Main Mode)
162	12:09:28.672144	1.1.1.1	2.2.2.2	ISAKMP	214	Quick Mode
163	12:09:28.926799	2.2.2.2	1.1.1.1	ISAKMP	134	Informational
164	12:09:29.855434	2.2.2.2	1.1.1.1	ISAKMP	210	Identity Protection (Main Mode)
166	12:09:30.042589	1.1.1.1	2.2.2.2	ISAKMP	150	Identity Protection (Main Mode)
167	12:09:30.176062	2.2.2.2	1.1.1.1	ISAKMP	390	Identity Protection (Main Mode)
169	12:09:30.341166	1.1.1.1	2.2.2.2	ISAKMP	410	Identity Protection (Main Mode)
170	12:09:30.892237	2.2.2.2	1.1.1.1	ISAKMP	118	Identity Protection (Main Mode)
171	12:09:31.046912	1.1.1.1	2.2.2.2	ISAKMP	118	Identity Protection (Main Mode)
172	12:09:31.297021	2.2.2.2	1.1.1.1	ISAKMP	230	Quick Mode
173	12:09:31.401622	1.1.1.1	2.2.2.2	ISAKMP	134	Informational
185	12:10:00.474533	2.2.2.2	1.1.1.1	ISAKMP	230	Quick Mode
186	12:10:00.568552	1.1.1.1	2.2.2.2	ISAKMP	134	Informational
191	12:10:13.892447	1.1.1.1	2.2.2.2	ISAKMP	214	Quick Mode
192	12:10:13.959083	2.2.2.2	1.1.1.1	ISAKMP	134	Informational
211	12:10:58.901471	2.2.2.2	1.1.1.1	ISAKMP	230	Quick Mode
212	12:10:59.105788	1.1.1.1	2.2.2.2	ISAKMP	134	Informational
213	12:10:59.466884	1.1.1.1	2.2.2.2	ISAKMP	214	Quick Mode
214	12:10:59.621211	2.2.2.2	1.1.1.1	ISAKMP	134	Informational

Figure 9: Failed IPSec negotiation due to parameter mismatch

2.6 ESP Tunnel Operation

The transform set was later changed to ESP using the command:

```
crypto ipsec transform-set myTSet esp-aes esp-sha-hmac
```

ESP provides confidentiality in addition to authentication and integrity.

Figure 10 shows an ESP-protected packet. Unlike AH, only the ESP header remains visible, while the original payload is encrypted and cannot be inspected in Wireshark.

No.	Time	Source	Destination	Protocol	Leng	Info
2	12:30:08.264215	2.2.2.2	1.1.1.1	ESP	166	ESP (SPI=0x6e0b4bc2)

```
> Frame 2: Packet, 166 bytes on wire (1328 bits), 166 bytes captured (1328 bits) on interface -, id 0
> Ethernet II, Src: ca:01:08:63:00:00 (ca:01:08:63:00:00), Dst: 0c:d6:9b:f8:00:00 (0c:d6:9b:f8:00:00)
> Internet Protocol Version 4, Src: 2.2.2.2, Dst: 1.1.1.1
v Encapsulating Security Payload
    ESP SPI: 0x6e0b4bc2 (1846234050)
    ESP Sequence: 21
```

Figure 10: ESP protected packet

Figure 11 shows ESP packets exchanged together with OSPF Hello packets during tunnel operation.

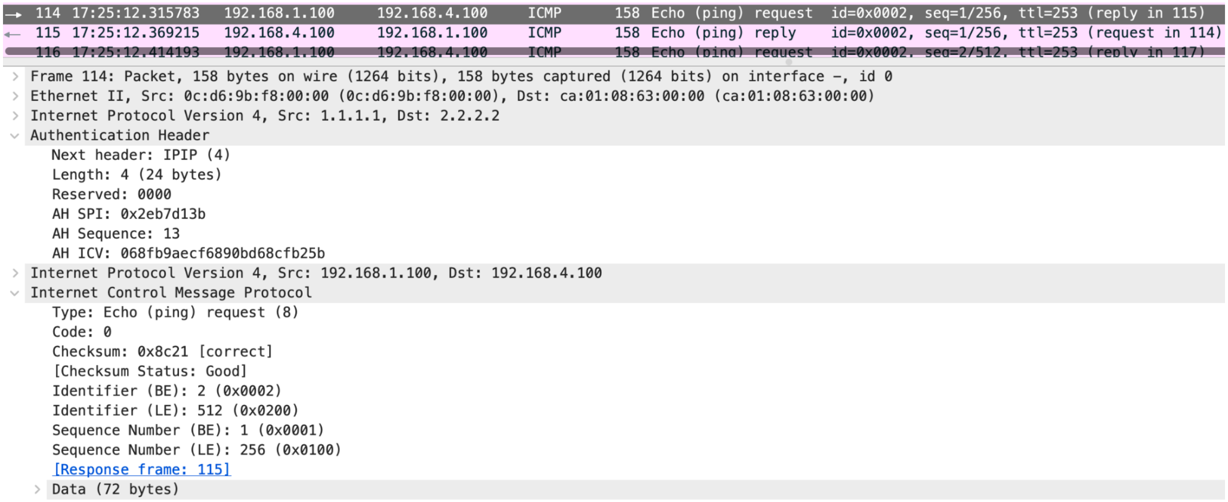


Figure 11: ESP traffic and OSPF exchanges

Compared to AH, ESP provides confidentiality by encrypting the payload, while AH only guarantees integrity and authentication. In AH mode, the original payload remains visible in Wireshark, whereas in ESP mode only the ESP header can be inspected and the payload contents remain encrypted.

2.7 Firewall Requirements for IPSec

In order for an IPSec tunnel to be successfully established through a firewall, the firewall must allow the protocols and ports used by IKE and IPSec.

The following traffic must be permitted:

- UDP port 500 (ISAKMP/IKE)
- ESP protocol (IP protocol 50)
- AH protocol (IP protocol 51), when AH is used

If NAT traversal is used, UDP port 4500 must also be allowed.

Blocking any of these protocols prevents successful establishment of the IPSec tunnel.

3 IKEv1 with Digital Signatures

3.1 Introduction

This section describes the configuration and analysis of IPSec VPN using IKEv1 with digital certificates (PKI-based authentication using SCEP). The objective is to replace pre-shared keys with RSA signatures issued by a Certification Authority (CA).

3.2 Topology Overview

RA operates as the Certification Authority (CA) and NTP server. R1 and R2 establish an IPSec tunnel authenticated using RSA signatures and certificates obtained through SCEP enrollment.

3.3 PKI Installation Verification

The Certification Authority (CA) and router certificates were verified using IOS show commands.

3.3.1 CA Configuration and Certificate (RA)

The CA is configured on router RA and operates in automatic enrollment mode.

```
RA#show crypto pki server
Certificate Server myCA:
    Status: enabled
    State: enabled
    Server's configuration is locked  (enter "shut" to unlock it)
    Issuer name: cn=saarCA
    CA cert fingerprint: 46CB9C10 CC370311 8CFCB545 E4DD2F58
    Granting mode is: auto
    Last certificate issued serial number (hex): 1
    CA certificate expiration timer: 16:17:23 UTC May 14 2029
    CRL NextUpdate timer: 22:17:23 UTC May 15 2026
    Current primary storage dir: nvram:
    Database Level: Minimum - no cert data written to storage
```

The CA certificate installed on RA confirms that RA is acting as a trusted root authority:

```
RA#show crypto ca certificate
CA Certificate
  Status: Available
  Certificate Serial Number (hex): 01
  Certificate Usage: Signature
  Issuer:
    cn=saarCA
  Subject:
```

```
    cn=saarCA
  Validity Date:
    start date: 16:17:23 UTC May 15 2026
    end   date: 16:17:23 UTC May 14 2029
  Associated Trustpoints: myCA
  Storage: nvram:saarCA#1CA.cer
```

The detailed certificate view confirms RSA key usage and X.509 extensions:

```
RA#show crypto ca certificate verbose
CA Certificate
  Status: Available
  Version: 3
  Certificate Serial Number (hex): 01
  Certificate Usage: Signature
  Issuer:
    cn=saarCA
  Subject:
    cn=saarCA
  Validity Date:
    start date: 16:17:23 UTC May 15 2026
    end   date: 16:17:23 UTC May 14 2029
  Subject Key Info:
    Public Key Algorithm: rsaEncryption
    RSA Public Key: (1024 bit)
  Signature Algorithm: MD5 with RSA Encryption
  Fingerprint MD5: 46CB9C10 CC370311 8CFCB545 E4DD2F58
  Fingerprint SHA1: 0B6BF47E F0932FC1 54A33FA8 4C1F48D1 80083958
  X509v3 extensions:
    X509v3 Key Usage: 86000000
      Digital Signature
      Key Cert Sign
      CRL Signature
    X509v3 Subject Key ID: 42BD5391 0D07D957 6CC26630 AABA09E4 A93162E2
    X509v3 Basic Constraints:
        CA: TRUE
    X509v3 Authority Key ID: 42BD5391 0D07D957 6CC26630 AABA09E4 A93162E2
    Authority Info Access:
  Associated Trustpoints: myCA
  Storage: nvram:saarCA#1CA.cer
```

3.3.2 Router Certificates (R1)

R1 successfully enrolled with the CA and received both the CA certificate and its own identity certificate:

```
R1#show crypto pki certificates
```

```
Certificate
  Status: Available
  Certificate Serial Number (hex): 03
  Certificate Usage: General Purpose
  Issuer:
    cn=saarCA
  Subject:
    Name: R1
    Serial Number: 9SO8DYF8WAKSOLHEG3ZVL
    hostname=R1+serialNumber=9SO8DYF8WAKSOLHEG3ZVL
  Validity Date:
    start date: 16:33:16 UTC May 15 2026
    end   date: 16:33:16 UTC May 15 2027
  Associated Trustpoints: myTrustpoint
```

The CA certificate is also stored locally to validate the trust chain:

```
CA Certificate
  Status: Available
  Certificate Serial Number (hex): 01
  Certificate Usage: Signature
  Issuer:
    cn=saarCA
  Subject:
    cn=saarCA
  Validity Date:
    start date: 16:17:23 UTC May 15 2026
    end   date: 16:17:23 UTC May 14 2029
  Associated Trustpoints: myTrustpoint
```

3.3.3 RSA Public Keys (R1 and R2)

The RSA public keys generated on R1 and R2 are extracted using the following IOS command:

```
show crypto key mypub rsa
```

R2 RSA Public Key:

```
R2#show crypto key mypub rsa
% Key pair was generated at: 16:32:05 UTC May 15 2026
Key name: R2
Key type: RSA KEYS
 Storage Device: private-config
 Usage: General Purpose Key
 Key is not exportable.
 Key Data:
```

```
  30819F30 0D06092A 864886F7 0D010101 05000381 8D003081 89028181 00C337FE
  B0DAEE4D 444B0AF7 A19A73B3 2D2FCD5E 27E09095 3017E393 2F27F7AA 581894E6
  5E7F50B3 0454ACF2 FD1E8E24 F91125DD 6D611D93 611D70AD 36C59F6F EAEE003C
  7704B6DB 41D9A165 27252864 17170DA2 F38A68D4 46C91133 36DBCBC5 B09AD90E
  0F05531B F7ABB4A5 0B954C0F 389C2195 3DE377FF 7FE63FB3 B02DF189 35020301
  0001
% Key pair was generated at: 16:57:46 UTC May 16 2026
Key name: R2.server
Key type: RSA KEYS
Temporary key
 Usage: Encryption Key
 Key is not exportable.
 Key Data:
  307C300D 06092A86 4886F70D 01010105 00036B00 30680261 00A38B5F B33F29E4
  E5E30AFB 512084C9 0A89E015 BCA1569A C10C686B B79BD8EC C3C25CAD 6EE74C8F
  C891E5FF 1B042503 710EC940 65712523 377F2776 77CDC4C0 D7D61754 57C88063
  19BF191B F48130CC 327EF733 1F877D95 5CB74450 AEEF7E99 45020301 0001
```

R1 RSA Public Key:

```
R1#show crypto key mypub rsa
% Key pair was generated at: 16:31:43 UTC May 15 2026
Key name: R1
Key type: RSA KEYS
 Storage Device: private-config
 Usage: General Purpose Key
 Key is not exportable.
 Key Data:
  30819F30 0D06092A 864886F7 0D010101 05000381 8D003081 89028181 00DF87BB
  11BD33EE C4E9DF4A C2FF2CCB 2ECCEFF9 713AAA45 633ADE4A C188109E 88680D67
  FB0A2F20 67257007 79DB3616 75D5459A 4898DE69 E74E80BA 77C993E8 A4B70D41
  00D1D2FB E37BF0E3 89C8BEEE 9830712E 8A4C13DC E30C5C72 D72FEDF2 A221F6DB
  9A1A1B8F 9A69BC61 3700CA1B B66DB044 72F24609 E0C1FE8D 24D1B3DA DD020301
  0001
% Key pair was generated at: 16:57:42 UTC May 16 2026
Key name: R1.server
Key type: RSA KEYS
Temporary key
 Usage: Encryption Key
 Key is not exportable.
 Key Data:
  307C300D 06092A86 4886F70D 01010105 00036B00 30680261 0095CC14 73EE9BA8
  1B0A40A6 E3EE90C1 BC2FAADF 2D532270 160D9039 C19C40CA 6A4E54B5 BAB8892C
  D266AE9A 53210F58 876755F6 381C44B6 5D71F57F 35705BC5 3E8D9746 61833A6A
  DB5E740B 8A22A486 83EE7AFA ACAB3798 27D03FC2 7C6EA875 FD020301 0001
```

Both routers generate 1024-bit RSA key pairs used for digital signature authentication during IKEv1 Phase 1.

3.3.4 Summary

The outputs confirm that:

- RA is correctly configured as a trusted CA (myCA).
- R1 has successfully obtained a signed X.509 certificate from the CA.
- The certificate chain is self-signed at the CA level.
- RSA key pairs are generated locally on each router and are bound to X.509 certificates issued by the CA for IKEv1 authentication using digital signatures.
- Both R1 and R2 possess locally generated RSA key pairs used for authentication and certificate binding in IKEv1.

3.4 SCEP Enrollment Analysis

SCEP (Simple Certificate Enrollment Protocol) is used to automate certificate distribution between routers and the CA using HTTP.

The enrollment process consists of:

- `CA certificate request`
- `CA certificate response`
- `Router certificate request`
- `Router certificate issuance`

3.4.1 CA Certificate Retrieval

Initially, R1 requests the CA certificate using the SCEP `GetCACert` operation over HTTP.

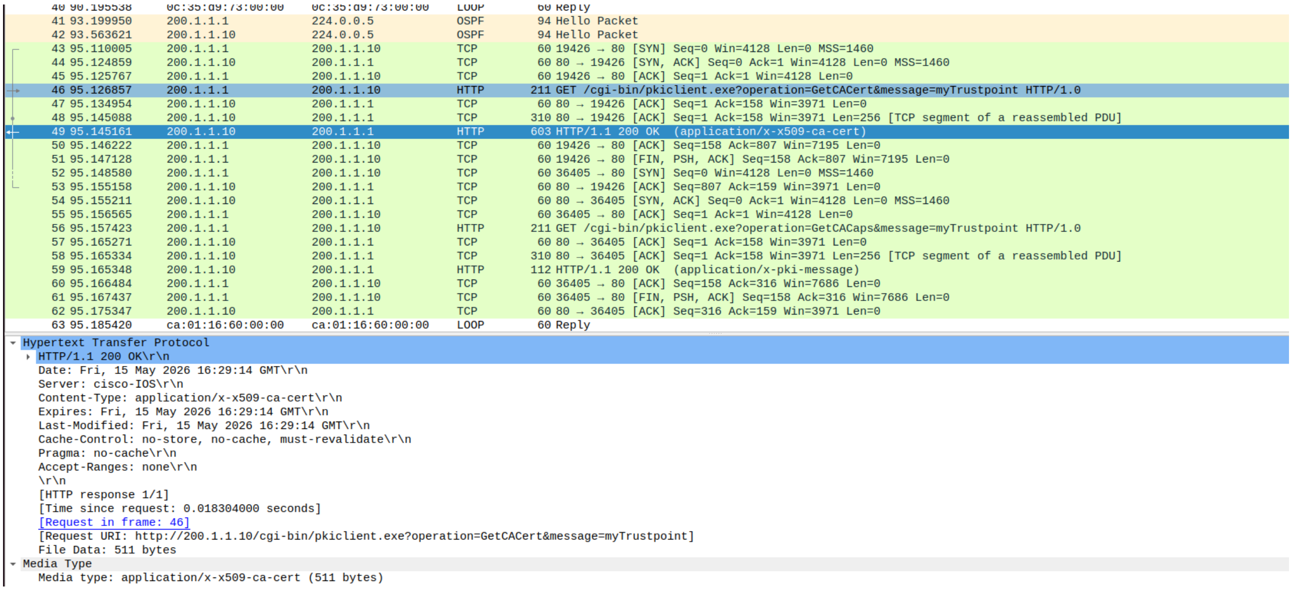


Figure 12: SCEP GetCACert request and CA certificate response

The CA replies with an HTTP `200 OK` message containing an X.509 CA certificate with media type `application/x-x509-ca-cert`. This certificate allows R1 to establish trust with the Certification Authority before certificate enrollment.

3.4.2 Certificate Enrollment Request

After establishing trust with the CA, R1 performs certificate enrollment using the SCEP `PKIOperation` request.

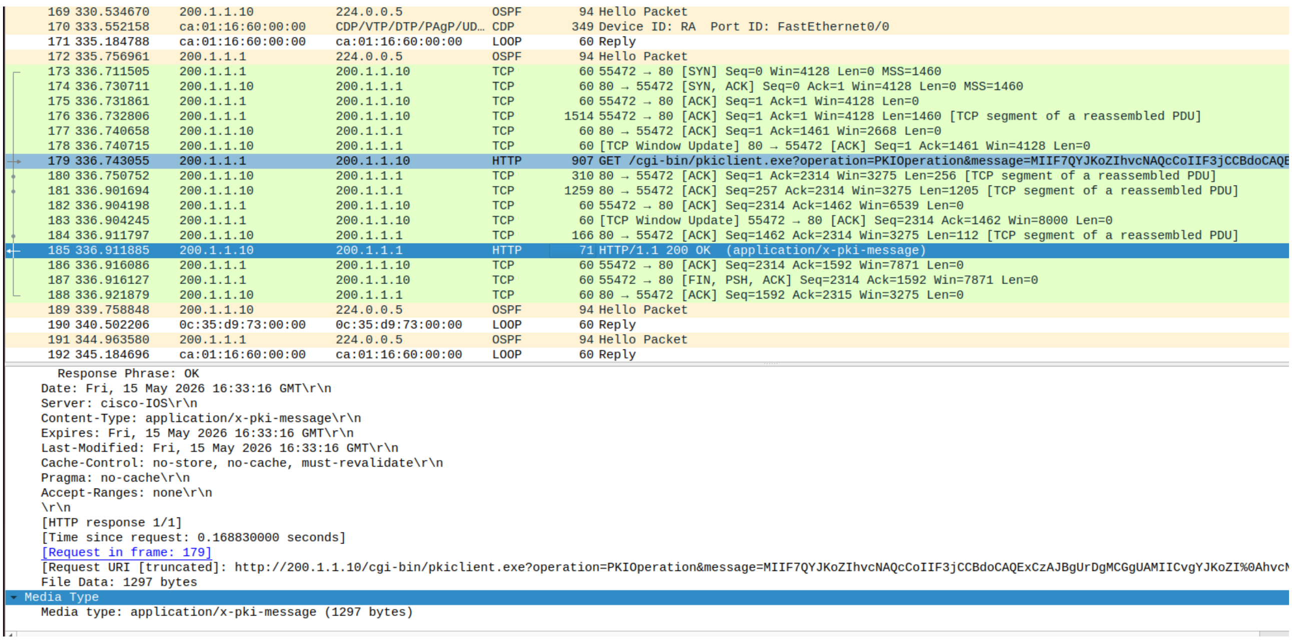


Figure 13: SCEP PKIOperation certificate enrollment request

The HTTP request contains a PKCS#10 certificate signing request (CSR) encoded in the message field. The CSR includes the router identity and its RSA public key, allowing the CA to generate a signed X.509 certificate for R1.

The captured HTTP messages show certificate exchange between R1 and the CA. The content-type field indicates certificate payload transfer (.crt).

Using Wireshark, the packet containing the issued certificate was identified by filtering HTTP traffic and locating the response message with a `Media Type` indicating a certificate object. The packet was then selected, and the certificate data was extracted by right-clicking on the relevant field and choosing *Export Packet Bytes*. The file was saved with a `.crt` extension, which allows it to be decoded as an X.509 certificate.

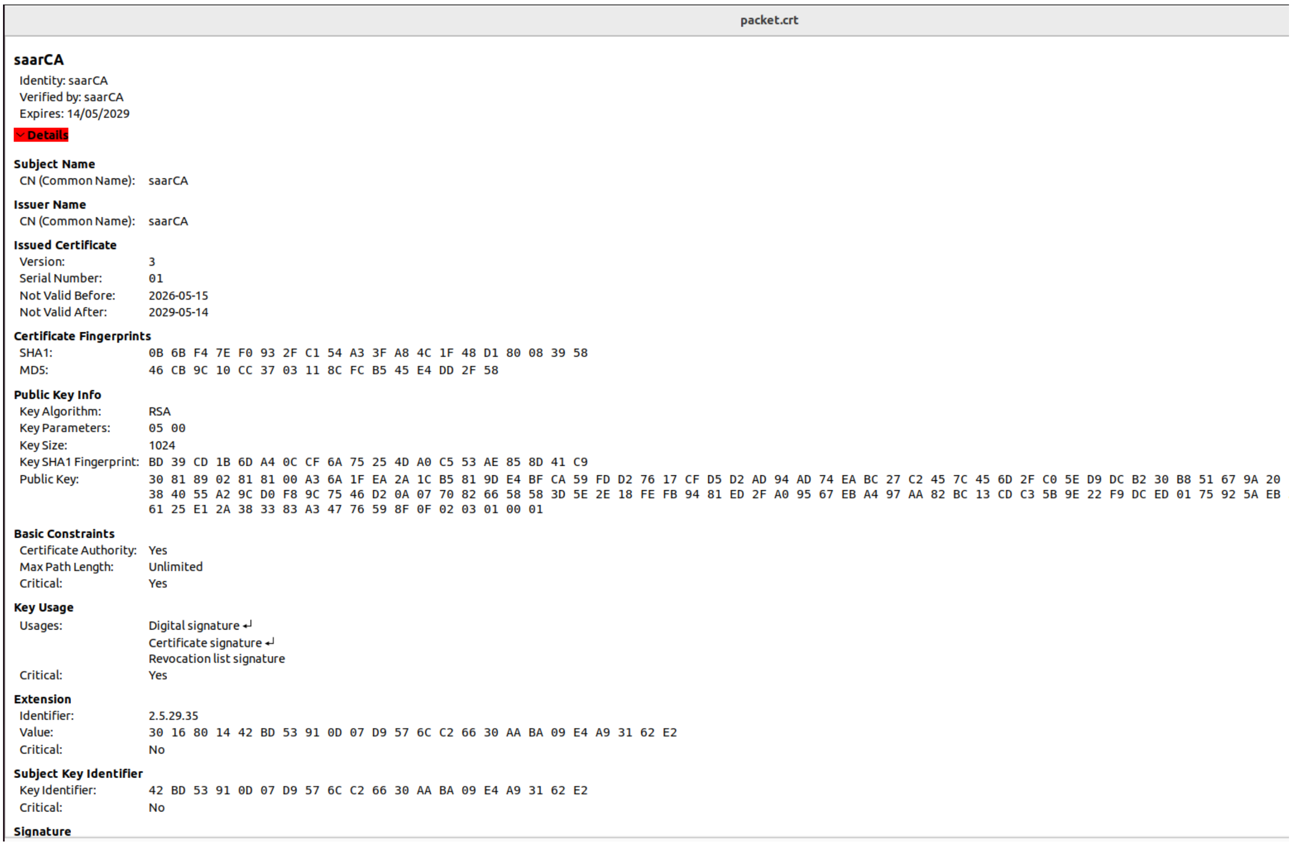


Figure 14: HTTP packet carrying the issued X.509 certificate (.crt file export in Wireshark)

The exported certificate represents the CA-signed identity certificate for R1. It contains the subject information, the public key corresponding to R1, and the digital signature generated by the CA, which validates its authenticity within the PKI infrastructure.

3.5 Encrypted ICMP Traffic Analysis

After configuring IPSec with RSA-signature authentication, connectivity between PC1 and PC2 was successfully verified using ICMP echo requests (ping).

Traffic was captured on the public network between R1 and R2 while the ping operation was performed.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:35:d9:73:00:00	0c:35:d9:73:00:00	LOOP	60	Reply
2	3.889135	200.1.1.10	224.0.0.5	OSPF	94	Hello Packet
3	5.482695	ca:01:16:60:00:00	ca:01:16:60:00:00	LOOP	60	Reply
4	6.320045	2.2.2.2	1.1.1.1	ESP	166	ESP (SPI=0xb87d2aab)
5	7.009428	1.1.1.1	2.2.2.2	ESP	166	ESP (SPI=0x147d646a)
6	7.269011	200.1.1.1	224.0.0.5	OSPF	94	Hello Packet
7	8.171183	2.2.2.2	1.1.1.1	ESP	182	ESP (SPI=0xb87d2aab)
8	8.188890	1.1.1.1	2.2.2.2	ESP	182	ESP (SPI=0x147d646a)
9	8.221515	2.2.2.2	1.1.1.1	ESP	182	ESP (SPI=0xb87d2aab)
10	8.239182	1.1.1.1	2.2.2.2	ESP	182	ESP (SPI=0x147d646a)
11	8.282182	2.2.2.2	1.1.1.1	ESP	182	ESP (SPI=0xb87d2aab)
12	8.299668	1.1.1.1	2.2.2.2	ESP	182	ESP (SPI=0x147d646a)
13	8.322552	2.2.2.2	1.1.1.1	ESP	182	ESP (SPI=0xb87d2aab)
14	8.339979	1.1.1.1	2.2.2.2	ESP	182	ESP (SPI=0x147d646a)
15	8.362850	2.2.2.2	1.1.1.1	ESP	182	ESP (SPI=0xb87d2aab)
16	8.380385	1.1.1.1	2.2.2.2	ESP	182	ESP (SPI=0x147d646a)
17	10.028466	0c:35:d9:73:00:00	0c:35:d9:73:00:00	LOOP	60	Reply
18	13.658516	200.1.1.10	224.0.0.5	OSPF	94	Hello Packet
19	15.485532	ca:01:16:60:00:00	ca:01:16:60:00:00	LOOP	60	Reply
20	15.847636	2.2.2.2	1.1.1.1	ESP	166	ESP (SPI=0xb87d2aab)
21	16.864505	200.1.1.1	224.0.0.5	OSPF	94	Hello Packet
22	17.007888	1.1.1.1	2.2.2.2	ESP	166	ESP (SPI=0x147d646a)
23	20.042916	0c:35:d9:73:00:00	0c:35:d9:73:00:00	LOOP	60	Reply

```
▸ Frame 7: 182 bytes on wire (1456 bits), 182 bytes captured (1456 bits) on interface -, id 0
▸ Ethernet II, Src: ca:01:16:60:00:00 (ca:01:16:60:00:00), Dst: 0c:35:d9:73:00:00 (0c:35:d9:73:00:00)
▾ Internet Protocol Version 4, Src: 2.2.2.2, Dst: 1.1.1.1
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  ▸ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 168
    Identification: 0x011d (285)
  ▸ Flags: 0x00
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 254
    Protocol: Encap Security Payload (50)
    Header Checksum: 0xb501 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 2.2.2.2
    Destination Address: 1.1.1.1
▾ Encapsulating Security Payload
    ESP SPI: 0xb87d2aab (3095210667)
    ESP Sequence: 61
```

Figure 15: Encrypted IPSec traffic observed between R1 and R2

The Wireshark capture shows that the ICMP packets exchanged between PC1 and PC2 are not visible in plaintext on the public network. Instead, the packets appear encapsulated using ESP (Encapsulating Security Payload).

The ESP packets contain encrypted payload data, preventing direct inspection of the original ICMP echo request and echo reply contents. Only the external IP header remains visible, showing communication between the tunnel endpoints (R1 and R2).

This confirms that IPSec is correctly providing:

- **Confidentiality** through payload encryption
- **Integrity protection** through ESP authentication
- **Secure tunnel encapsulation** between the two sites

The observed ESP traffic corresponds to the IPSec transform-set configuration:

```
crypto ipsec transform-set myTSet esp-aes esp-sha-hmac
```

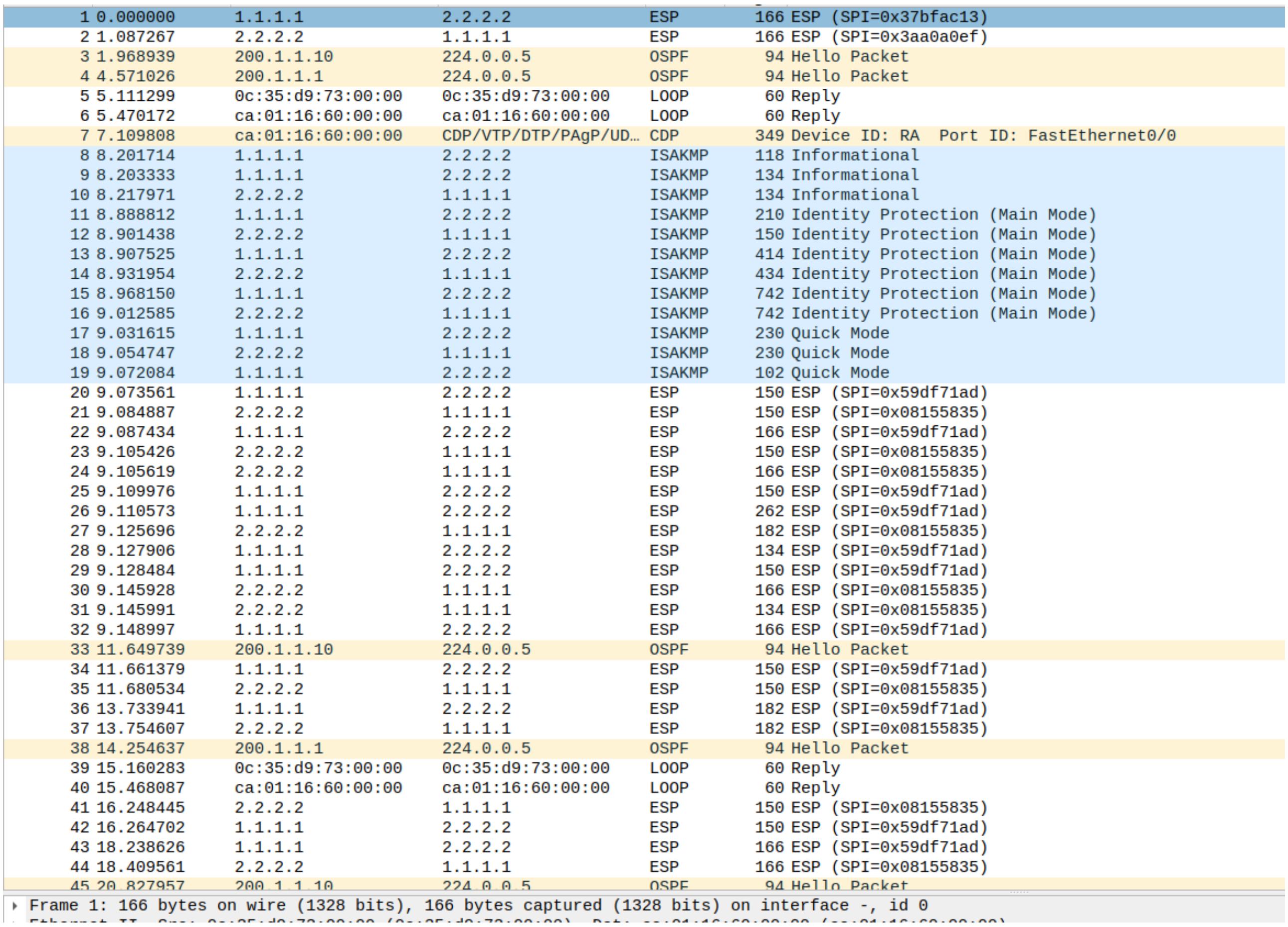
AES is used for encryption while SHA-HMAC is used for integrity verification.

3.6 ISAKMP and SA Re-establishment Analysis

To analyze tunnel re-establishment, the IPSec Security Associations (SAs) were manually cleared using:

```
clear crypto session
```

While capturing traffic on the public network, new ISAKMP exchanges were observed as the routers negotiated fresh IPSec Security Associations.



1	0.000000	1.1.1.1	2.2.2.2	ESP	166	ESP (SPI=0x37bfac13)
2	1.087267	2.2.2.2	1.1.1.1	ESP	166	ESP (SPI=0x3aa0a0ef)
3	1.968939	200.1.1.10	224.0.0.5	OSPF	94	Hello Packet
4	4.571026	200.1.1.1	224.0.0.5	OSPF	94	Hello Packet
5	5.111299	0c:35:d9:73:00:00	0c:35:d9:73:00:00	LOOP	60	Reply
6	5.470172	ca:01:16:60:00:00	ca:01:16:60:00:00	LOOP	60	Reply
7	7.109808	ca:01:16:60:00:00	CDP/VTP/DTP/PAgP/UD…	CDP	349	Device ID: RA Port ID: FastEthernet0/0
8	8.201714	1.1.1.1	2.2.2.2	ISAKMP	118	Informational
9	8.203333	1.1.1.1	2.2.2.2	ISAKMP	134	Informational
10	8.217971	2.2.2.2	1.1.1.1	ISAKMP	134	Informational
11	8.888812	1.1.1.1	2.2.2.2	ISAKMP	210	Identity Protection (Main Mode)
12	8.901438	2.2.2.2	1.1.1.1	ISAKMP	150	Identity Protection (Main Mode)
13	8.907525	1.1.1.1	2.2.2.2	ISAKMP	414	Identity Protection (Main Mode)
14	8.931954	2.2.2.2	1.1.1.1	ISAKMP	434	Identity Protection (Main Mode)
15	8.968150	1.1.1.1	2.2.2.2	ISAKMP	742	Identity Protection (Main Mode)
16	9.012585	2.2.2.2	1.1.1.1	ISAKMP	742	Identity Protection (Main Mode)
17	9.031615	1.1.1.1	2.2.2.2	ISAKMP	230	Quick Mode
18	9.054747	2.2.2.2	1.1.1.1	ISAKMP	230	Quick Mode
19	9.072084	1.1.1.1	2.2.2.2	ISAKMP	102	Quick Mode
20	9.073561	1.1.1.1	2.2.2.2	ESP	150	ESP (SPI=0x59df71ad)
21	9.084887	2.2.2.2	1.1.1.1	ESP	150	ESP (SPI=0x08155835)
22	9.087434	1.1.1.1	2.2.2.2	ESP	166	ESP (SPI=0x59df71ad)
23	9.105426	2.2.2.2	1.1.1.1	ESP	150	ESP (SPI=0x08155835)
24	9.105619	2.2.2.2	1.1.1.1	ESP	166	ESP (SPI=0x08155835)
25	9.109976	1.1.1.1	2.2.2.2	ESP	150	ESP (SPI=0x59df71ad)
26	9.110573	1.1.1.1	2.2.2.2	ESP	262	ESP (SPI=0x59df71ad)
27	9.125696	2.2.2.2	1.1.1.1	ESP	182	ESP (SPI=0x08155835)
28	9.127906	1.1.1.1	2.2.2.2	ESP	134	ESP (SPI=0x59df71ad)
29	9.128484	1.1.1.1	2.2.2.2	ESP	150	ESP (SPI=0x59df71ad)
30	9.145928	2.2.2.2	1.1.1.1	ESP	166	ESP (SPI=0x08155835)
31	9.145991	2.2.2.2	1.1.1.1	ESP	134	ESP (SPI=0x08155835)
32	9.148997	1.1.1.1	2.2.2.2	ESP	166	ESP (SPI=0x59df71ad)
33	11.649739	200.1.1.10	224.0.0.5	OSPF	94	Hello Packet
34	11.661379	1.1.1.1	2.2.2.2	ESP	150	ESP (SPI=0x59df71ad)
35	11.680534	2.2.2.2	1.1.1.1	ESP	150	ESP (SPI=0x08155835)
36	13.733941	1.1.1.1	2.2.2.2	ESP	182	ESP (SPI=0x59df71ad)
37	13.754607	2.2.2.2	1.1.1.1	ESP	182	ESP (SPI=0x08155835)
38	14.254637	200.1.1.1	224.0.0.5	OSPF	94	Hello Packet
39	15.160283	0c:35:d9:73:00:00	0c:35:d9:73:00:00	LOOP	60	Reply
40	15.468087	ca:01:16:60:00:00	ca:01:16:60:00:00	LOOP	60	Reply
41	16.248445	2.2.2.2	1.1.1.1	ESP	150	ESP (SPI=0x08155835)
42	16.264702	1.1.1.1	2.2.2.2	ESP	150	ESP (SPI=0x59df71ad)
43	18.238626	1.1.1.1	2.2.2.2	ESP	166	ESP (SPI=0x59df71ad)
44	18.409561	2.2.2.2	1.1.1.1	ESP	166	ESP (SPI=0x08155835)
45	20.827957	200.1.1.10	224.0.0.5	OSPF	94	Hello Packet

Frame 1: 166 bytes on wire (1328 bits), 166 bytes captured (1328 bits) on interface -, id 0

Figure 16: ISAKMP exchanges after clearing crypto sessions

After the existing SAs were deleted, R1 and R2 initiated a new IKEv1 Phase 1 negotiation using ISAKMP messages over UDP port 500.

The captured exchange shows the following operations:

- ISAKMP SA proposal negotiation
- Exchange of nonces and Diffie-Hellman values
- Certificate exchange between peers

- RSA digital signature authentication
- Establishment of a new ISAKMP Security Association
- Negotiation of IPSec Phase 2 parameters
- Creation of new IPSec SAs for ESP traffic

The certificate-based authentication process relies on the X.509 certificates previously obtained from the CA through SCEP enrollment.

During authentication, each router validates:

- The peer certificate authenticity
- The CA signature
- The RSA signature generated during IKE authentication

Successful validation allows the tunnel to be securely re-established without the use of pre-shared keys.

The re-established IPSec tunnel was confirmed by successful encrypted ICMP communication between PC1 and PC2 after the negotiation process completed.

3.7 Peer Public Key Storage Verification

After successful IKEv1 authentication and IPSec tunnel establishment, each router stores the public key information of the remote peer.

At R1, the public key of R2 can be viewed using:

```
show crypto key pubkey-chain rsa name R2
```

The output confirms that R1 has learned and stored the RSA public key associated with R2's certificate.

```
R1#sh crypto key pubkey-chain rsa name R2
 Key name: R2
 Subject name(X.500 DN name): hostname=R2+serialNumber=9C92ZG7WMXGLWHR7XWAK8
 Key id: 11
 Serial number: 02
 Usage: Signature Key
 Source: Certificate
 Data:
  30819F30 0D06092A 864886F7 0D010101 05000381 8D003081 89028181 00C337FE
  B0DAEE4D 444B0AF7 A19A73B3 2D2FCD5E 27E09095 3017E393 2F27F7AA 581894E6
  5E7F50B3 0454ACF2 FD1E8E24 F91125DD 6D611D93 611D70AD 36C59F6F EAEE003C
  7704B6DB 41D9A165 27252864 17170DA2 F38A68D4 46C91133 36DBCBC5 B09AD90E
  0F05531B F7ABB4A5 0B954C0F 389C2195 3DE377FF 7FE63FB3 B02DF189 35020301
  0001
```

This public key is extracted from the X.509 certificate presented by R2 during the IKEv1 authentication process.

The stored public key allows R1 to:

- Verify digital signatures generated by R2
- Validate the identity of the remote peer
- Establish trusted IPSec communications

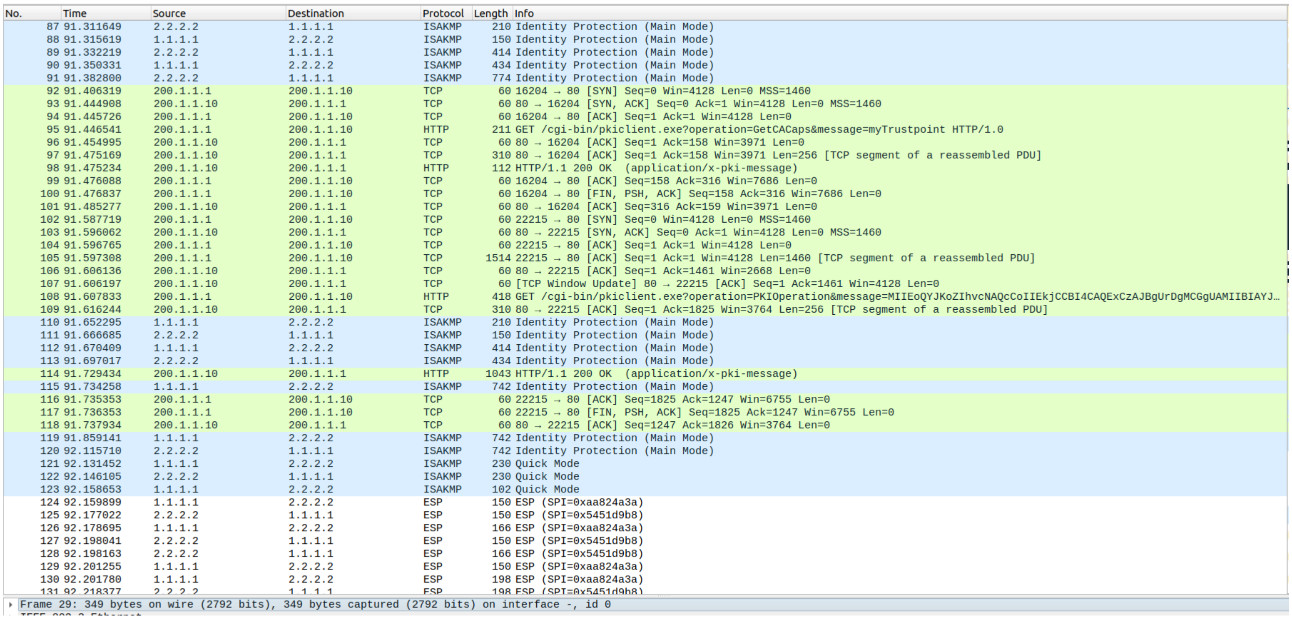
Similarly, R2 stores the public key of R1 after successful authentication.

The existence of the peer public key chain confirms that certificate-based authentication using RSA signatures was successfully completed and that both routers trust certificates issued by the Certification Authority.

3.8 SCEP Messages Observed After Router Restart

After stopping and restarting all routers, Wireshark captures were collected on the public network shortly after boot.

During startup, HTTP messages related to SCEP were again exchanged between the routers and the CA.



No.	Time	Source	Destination	Protocol	Length	Info
87	91.311649	2.2.2.2	1.1.1.1	ISAKMP	210	Identity Protection (Main Mode)
88	91.315619	1.1.1.1	2.2.2.2	ISAKMP	150	Identity Protection (Main Mode)
89	91.332219	2.2.2.2	1.1.1.1	ISAKMP	414	Identity Protection (Main Mode)
90	91.350331	1.1.1.1	2.2.2.2	ISAKMP	434	Identity Protection (Main Mode)
91	91.382800	2.2.2.2	1.1.1.1	ISAKMP	774	Identity Protection (Main Mode)
92	91.406319	200.1.1.1	200.1.1.10	TCP	60	16204 → 80 [SYN] Seq=0 Win=4128 Len=0 MSS=1460
93	91.444908	200.1.1.10	200.1.1.1	TCP	60	80 → 16204 [SYN, ACK] Seq=0 Ack=1 Win=4128 Len=0 MSS=1460
94	91.445726	200.1.1.1	200.1.1.10	TCP	60	16204 → 80 [ACK] Seq=1 Ack=1 Win=4128 Len=0
95	91.446541	200.1.1.1	200.1.1.10	HTTP	211	GET /cgi-bin/pkiclient.exe?operation=GetCACaps&message=myTrustpoint HTTP/1.0
96	91.454995	200.1.1.10	200.1.1.1	TCP	60	80 → 16204 [ACK] Seq=1 Ack=158 Win=3971 Len=0
97	91.475169	200.1.1.10	200.1.1.1	TCP	310	80 → 16204 [ACK] Seq=1 Ack=158 Win=3971 Len=256 [TCP segment of a reassembled PDU]
98	91.475234	200.1.1.10	200.1.1.1	HTTP	112	HTTP/1.1 200 OK (application/x-pki-message)
99	91.476088	200.1.1.1	200.1.1.10	TCP	60	16204 → 80 [ACK] Seq=158 Ack=316 Win=7686 Len=0
100	91.476837	200.1.1.1	200.1.1.10	TCP	60	16204 → 80 [FIN, PSH, ACK] Seq=158 Ack=316 Win=7686 Len=0
101	91.485277	200.1.1.10	200.1.1.1	TCP	60	80 → 16204 [ACK] Seq=316 Ack=159 Win=3971 Len=0
102	91.587719	200.1.1.1	200.1.1.10	TCP	60	22215 → 80 [SYN] Seq=0 Win=4128 Len=0 MSS=1460
103	91.596062	200.1.1.10	200.1.1.1	TCP	60	80 → 22215 [SYN, ACK] Seq=0 Ack=1 Win=4128 Len=0 MSS=1460
104	91.596765	200.1.1.1	200.1.1.10	TCP	60	22215 → 80 [ACK] Seq=1 Ack=1 Win=4128 Len=0
105	91.597308	200.1.1.1	200.1.1.10	TCP	1514	22215 → 80 [ACK] Seq=1 Ack=1 Win=4128 Len=1460 [TCP segment of a reassembled PDU]
106	91.606136	200.1.1.10	200.1.1.1	TCP	60	80 → 22215 [ACK] Seq=1 Ack=1461 Win=2668 Len=0
107	91.606197	200.1.1.10	200.1.1.1	TCP	60	[TCP Window Update] 80 → 22215 [ACK] Seq=1 Ack=1461 Win=4128 Len=0
108	91.607833	200.1.1.1	200.1.1.10	HTTP	418	GET /cgi-bin/pkiclient.exe?operation=PKIOperation&message=MIIEoQYJKoZIhvcNAQcCoIIEkjCCBI4CAQExCzAJBgUrDgMCGgUAMIIBIAYJ…
109	91.616244	200.1.1.10	200.1.1.1	TCP	310	80 → 22215 [ACK] Seq=1 Ack=1825 Win=3764 Len=256 [TCP segment of a reassembled PDU]
110	91.652295	1.1.1.1	2.2.2.2	ISAKMP	210	Identity Protection (Main Mode)
111	91.666685	2.2.2.2	1.1.1.1	ISAKMP	150	Identity Protection (Main Mode)
112	91.670409	1.1.1.1	2.2.2.2	ISAKMP	414	Identity Protection (Main Mode)
113	91.697017	2.2.2.2	1.1.1.1	ISAKMP	434	Identity Protection (Main Mode)
114	91.729434	200.1.1.10	200.1.1.1	HTTP	1043	HTTP/1.1 200 OK (application/x-pki-message)
115	91.734258	1.1.1.1	2.2.2.2	ISAKMP	742	Identity Protection (Main Mode)
116	91.735353	200.1.1.1	200.1.1.10	TCP	60	22215 → 80 [ACK] Seq=1825 Ack=1247 Win=6755 Len=0
117	91.736353	200.1.1.1	200.1.1.10	TCP	60	22215 → 80 [FIN, PSH, ACK] Seq=1825 Ack=1247 Win=6755 Len=0
118	91.737934	200.1.1.10	200.1.1.1	TCP	60	80 → 22215 [ACK] Seq=1247 Ack=1826 Win=3764 Len=0
119	91.859141	1.1.1.1	2.2.2.2	ISAKMP	742	Identity Protection (Main Mode)
120	92.115710	2.2.2.2	1.1.1.1	ISAKMP	742	Identity Protection (Main Mode)
121	92.131452	1.1.1.1	2.2.2.2	ISAKMP	230	Quick Mode
122	92.146105	2.2.2.2	1.1.1.1	ISAKMP	230	Quick Mode
123	92.158653	1.1.1.1	2.2.2.2	ISAKMP	102	Quick Mode
124	92.159899	1.1.1.1	2.2.2.2	ESP	150	ESP (SPI=0xaa824a3a)
125	92.177022	2.2.2.2	1.1.1.1	ESP	150	ESP (SPI=0x5451d9b8)
126	92.178695	1.1.1.1	2.2.2.2	ESP	166	ESP (SPI=0xaa824a3a)
127	92.198041	2.2.2.2	1.1.1.1	ESP	150	ESP (SPI=0x5451d9b8)
128	92.198163	2.2.2.2	1.1.1.1	ESP	166	ESP (SPI=0x5451d9b8)
129	92.201255	1.1.1.1	2.2.2.2	ESP	150	ESP (SPI=0xaa824a3a)
130	92.201780	1.1.1.1	2.2.2.2	ESP	198	ESP (SPI=0xaa824a3a)
131	92.218377	2.2.2.2	1.1.1.1	ESP	198	ESP (SPI=0x5451d9b8)

Frame 29: 349 bytes on wire (2792 bits), 349 bytes captured (2792 bits) on interface -, id 0

Figure 17: SCEP-related HTTP traffic observed after router restart

The captured traffic includes HTTP requests and responses associated with certificate validation and certificate status retrieval operations.

These SCEP exchanges occur because the routers must re-establish trust information after reboot and verify the validity of the installed certificates before using them for IPSec authentication.

The observed messages serve several purposes:

- Verification that the CA is reachable
- Retrieval or validation of CA information
- Validation of certificate status
- Synchronization of trust-related information

This mechanism ensures that routers do not use invalid, expired, or revoked certificates during IKEv1 authentication.

The SCEP communication therefore plays an essential role in maintaining the integrity and trustworthiness of the PKI infrastructure used for IPSec VPN authentication.

3.9 Certificate Revocation and Authentication Failure

The certificates issued to R1 and R2 were revoked at the Certification Authority using:

```
RA#crypto pki server myCA revoke 02
RA#crypto pki server myCA revoke 03
```

After revocation, all routers were restarted and Wireshark captures were collected on the links between the routers and the CA.

When attempting communication between PC1 and PC2, the ping operation failed and the IPSec tunnel could not be established.

3.9.1 SCEP and Certificate Validation Traffic

After reboot, HTTP/SCEP-related messages were observed between the routers and the CA.

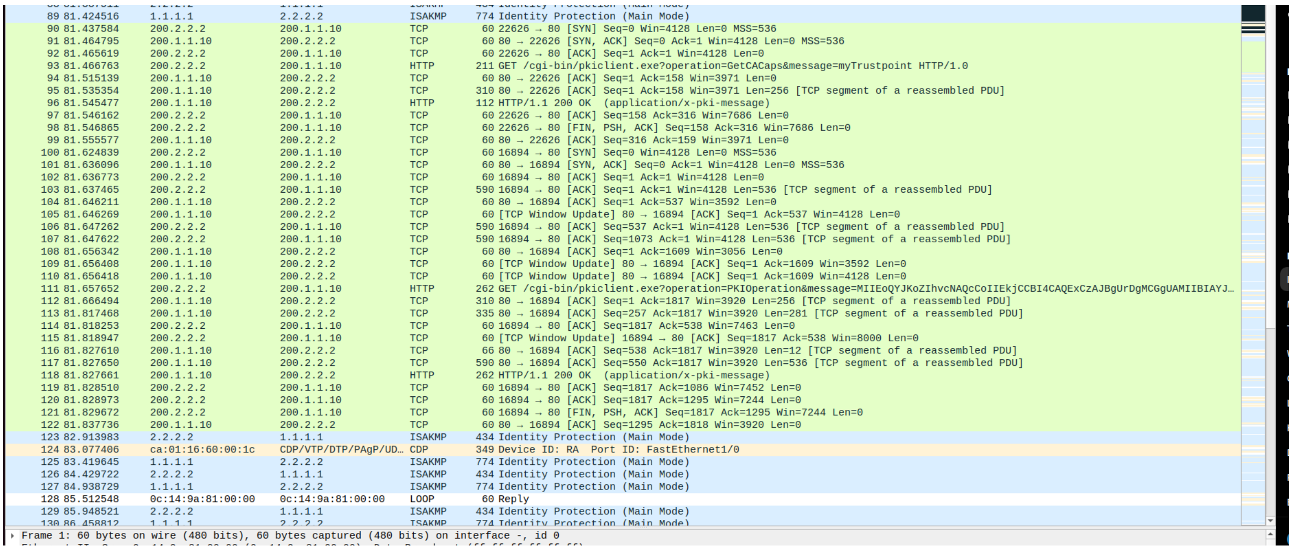


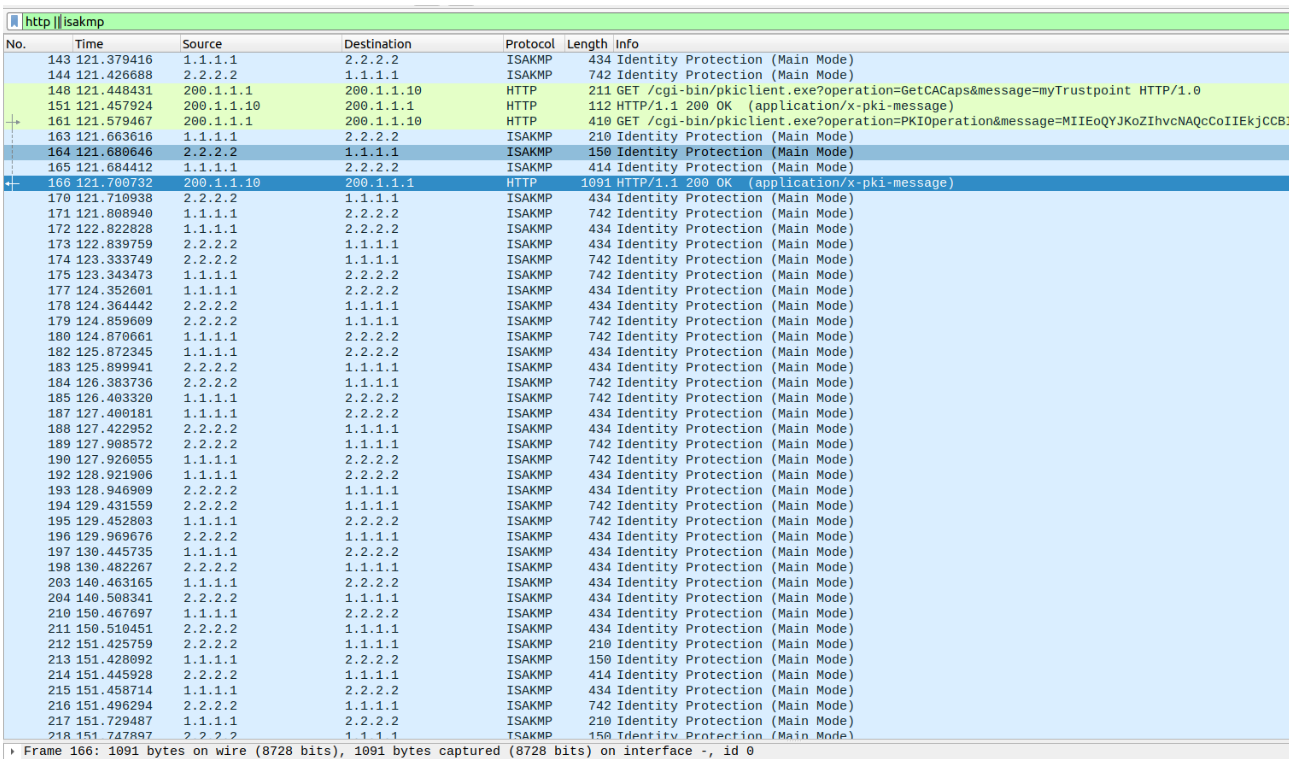
Figure 18: SCEP-related traffic observed after certificate revocation

These messages are associated with certificate validation and trust verification operations performed before IPSec authentication.

The routers contact the CA to verify the validity of certificates and retrieve updated revocation information.

3.9.2 Failed ISAKMP Negotiation

While the routers attempted to establish the IPSec tunnel, repeated ISAKMP exchanges were observed.



http || isakmp

No.	Time	Source	Destination	Protocol	Length	Info
143	121.379416	1.1.1.1	2.2.2.2	ISAKMP	434	Identity Protection (Main Mode)
144	121.426688	2.2.2.2	1.1.1.1	ISAKMP	742	Identity Protection (Main Mode)
148	121.448431	200.1.1.1	200.1.1.10	HTTP	211	GET /cgi-bin/pkiclient.exe?operation=GetCACaps&message=myTrustpoint HTTP/1.0
151	121.457924	200.1.1.10	200.1.1.1	HTTP	112	HTTP/1.1 200 OK (application/x-pki-message)
161	121.579467	200.1.1.1	200.1.1.10	HTTP	410	GET /cgi-bin/pkiclient.exe?operation=PKIOperation&message=MIIEoQYJKoZIhvcNAQcCoIIEkjCCBI
163	121.663616	1.1.1.1	2.2.2.2	ISAKMP	210	Identity Protection (Main Mode)
164	121.680646	2.2.2.2	1.1.1.1	ISAKMP	150	Identity Protection (Main Mode)
165	121.684412	1.1.1.1	2.2.2.2	ISAKMP	414	Identity Protection (Main Mode)
166	121.700732	200.1.1.10	200.1.1.1	HTTP	1091	HTTP/1.1 200 OK (application/x-pki-message)
170	121.710938	2.2.2.2	1.1.1.1	ISAKMP	434	Identity Protection (Main Mode)
171	121.808940	1.1.1.1	2.2.2.2	ISAKMP	742	Identity Protection (Main Mode)
172	122.822828	1.1.1.1	2.2.2.2	ISAKMP	434	Identity Protection (Main Mode)
173	122.839759	2.2.2.2	1.1.1.1	ISAKMP	434	Identity Protection (Main Mode)
174	123.333749	2.2.2.2	1.1.1.1	ISAKMP	742	Identity Protection (Main Mode)
175	123.343473	1.1.1.1	2.2.2.2	ISAKMP	742	Identity Protection (Main Mode)
177	124.352601	1.1.1.1	2.2.2.2	ISAKMP	434	Identity Protection (Main Mode)
178	124.364442	2.2.2.2	1.1.1.1	ISAKMP	434	Identity Protection (Main Mode)
179	124.859609	2.2.2.2	1.1.1.1	ISAKMP	742	Identity Protection (Main Mode)
180	124.870661	1.1.1.1	2.2.2.2	ISAKMP	742	Identity Protection (Main Mode)
182	125.872345	1.1.1.1	2.2.2.2	ISAKMP	434	Identity Protection (Main Mode)
183	125.899941	2.2.2.2	1.1.1.1	ISAKMP	434	Identity Protection (Main Mode)
184	126.383736	2.2.2.2	1.1.1.1	ISAKMP	742	Identity Protection (Main Mode)
185	126.403320	1.1.1.1	2.2.2.2	ISAKMP	742	Identity Protection (Main Mode)
187	127.400181	1.1.1.1	2.2.2.2	ISAKMP	434	Identity Protection (Main Mode)
188	127.422952	2.2.2.2	1.1.1.1	ISAKMP	434	Identity Protection (Main Mode)
189	127.908572	2.2.2.2	1.1.1.1	ISAKMP	742	Identity Protection (Main Mode)
190	127.926055	1.1.1.1	2.2.2.2	ISAKMP	742	Identity Protection (Main Mode)
192	128.921906	1.1.1.1	2.2.2.2	ISAKMP	434	Identity Protection (Main Mode)
193	128.946909	2.2.2.2	1.1.1.1	ISAKMP	434	Identity Protection (Main Mode)
194	129.431559	2.2.2.2	1.1.1.1	ISAKMP	742	Identity Protection (Main Mode)
195	129.452803	1.1.1.1	2.2.2.2	ISAKMP	742	Identity Protection (Main Mode)
196	129.969676	2.2.2.2	1.1.1.1	ISAKMP	434	Identity Protection (Main Mode)
197	130.445735	1.1.1.1	2.2.2.2	ISAKMP	434	Identity Protection (Main Mode)
198	130.482267	2.2.2.2	1.1.1.1	ISAKMP	434	Identity Protection (Main Mode)
203	140.463165	1.1.1.1	2.2.2.2	ISAKMP	434	Identity Protection (Main Mode)
204	140.508341	2.2.2.2	1.1.1.1	ISAKMP	434	Identity Protection (Main Mode)
210	150.467697	1.1.1.1	2.2.2.2	ISAKMP	434	Identity Protection (Main Mode)
211	150.510451	2.2.2.2	1.1.1.1	ISAKMP	434	Identity Protection (Main Mode)
212	151.425759	2.2.2.2	1.1.1.1	ISAKMP	210	Identity Protection (Main Mode)
213	151.428092	1.1.1.1	2.2.2.2	ISAKMP	150	Identity Protection (Main Mode)
214	151.445928	2.2.2.2	1.1.1.1	ISAKMP	414	Identity Protection (Main Mode)
215	151.458714	1.1.1.1	2.2.2.2	ISAKMP	434	Identity Protection (Main Mode)
216	151.496294	2.2.2.2	1.1.1.1	ISAKMP	742	Identity Protection (Main Mode)
217	151.729487	1.1.1.1	2.2.2.2	ISAKMP	210	Identity Protection (Main Mode)
218	151.747897	2.2.2.2	1.1.1.1	ISAKMP	150	Identity Protection (Main Mode)

Frame 166: 1091 bytes on wire (8728 bits), 1091 bytes captured (8728 bits) on interface -, id 0

Figure 19: Repeated ISAKMP negotiation attempts after certificate revocation

The repeated ISAKMP packets indicate that the routers continuously attempt to establish an IKEv1 Security Association but fail during the authentication phase.

This behavior was confirmed using IOS verification commands.

```
R1#sh crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst src state conn-id status
1.1.1.1 2.2.2.2 MM_KEY_EXCH 1040 ACTIVE
1.1.1.1 2.2.2.2 MM_KEY_EXCH 1038 ACTIVE
1.1.1.1 2.2.2.2 MM_NO_STATE 1036 ACTIVE (deleted)
1.1.1.1 2.2.2.2 MM_NO_STATE 1034 ACTIVE (deleted)
2.2.2.2 1.1.1.1 MM_KEY_EXCH 1041 ACTIVE
2.2.2.2 1.1.1.1 MM_NO_STATE 1039 ACTIVE (deleted)
2.2.2.2 1.1.1.1 MM_NO_STATE 1037 ACTIVE (deleted)

IPv6 Crypto ISAKMP SA
```

The ISAKMP state remains in `MM_KEY_EXCH`, indicating that Main Mode negotiation does not complete successfully.

A successfully established tunnel would normally reach the `QM_IDLE` state.

3.9.3 Absence of IPSec Security Associations

The IPSec SA database confirms that no operational IPSec tunnel was created.

```
R1#sh crypto ipsec sa

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 1.1.1.1

  protected vrf: (none)
  local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
  remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
  current_peer 2.2.2.2 port 500
   PERMIT, flags={origin_is_acl,} #pkts encaps: 0,
  #pkts encrypt: 0, #pkts digest: 0 #pkts decaps: 0,
  #pkts decrypt: 0, #pkts verify: 0 #pkts compressed: 0,
  #pkts decompressed: 0 #pkts not compressed: 0,
  #pkts compr. failed: 0 #pkts not decompressed: 0,
  #pkts decompress failed: 0
  #send errors 0, #recv errors 0

  local crypto endpt.: 1.1.1.1, remote crypto endpt.: 2.2.2.2
  plaintext mtu 1500, path mtu 1500, ip mtu 1500, ip mtu idb
GigabitEthernet0/0
  current outbound spi: 0x0(0)
  PFS (Y/N): N, DH group: none

  inbound esp sas:

  inbound ah sas:

  inbound pcp sas:

  outbound esp sas:

  outbound ah sas:

  outbound pcp sas:

R1#
```

The counters remain at zero because IPSec encryption was never activated.

Additionally, no ESP Security Associations were present:

```
inbound esp sas:
outbound esp sas:
```

This confirms that Phase 2 IPSec negotiation could not be completed.

3.9.4 Explanation of Communication Failure

Communication fails because the certificates used by R1 and R2 were revoked by the Certification Authority.

After reboot, the routers verify certificate validity during IKEv1 authentication. The CA indicates that the peer certificates are no longer trusted because they are listed in the Certificate Revocation List (CRL).

As a result:

- RSA signature authentication fails
- IKEv1 Main Mode negotiation cannot complete
- ISAKMP Security Associations are not successfully established
- IPSec Security Associations are never created
- ESP encrypted traffic cannot be generated
- ICMP communication between PC1 and PC2 fails

The experiment demonstrates the importance of certificate revocation in PKI-based IPSec infrastructures, ensuring that compromised or invalid certificates cannot be used to establish trusted VPN connections.

4 IPSec with IKEv2

4.1 Introduction

This section describes the configuration and analysis of an IPSec VPN using IKEv2 authentication with pre-shared keys. The objective is to establish a protected tunnel between routers R1 and R2 through the intermediate router RA and analyze the establishment and management of IKEv2 and IPSec Security Associations (SAs).

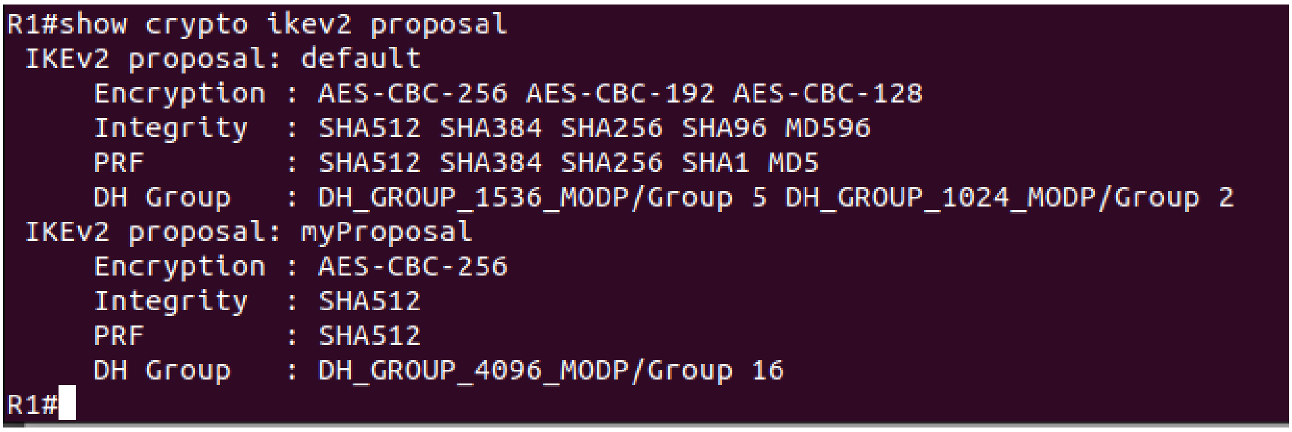
4.2 IKEv2 and IPSec Configuration

Both routers were configured with:

- an IKEv2 proposal using AES-CBC-256 encryption;
- SHA512 integrity and pseudo-random functions;
- Diffie-Hellman Group 16;

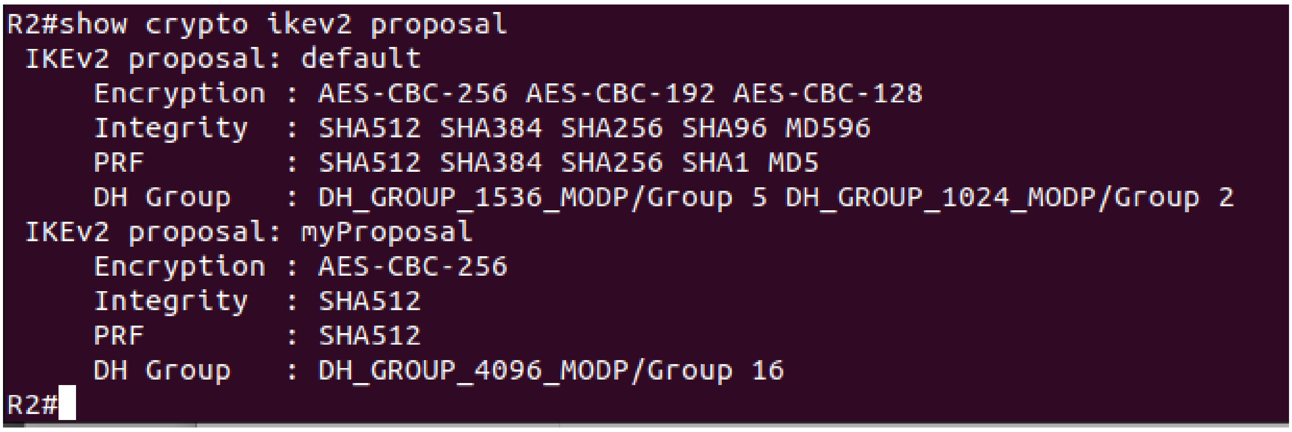
- an IPSec transform set using ESP-AES and ESP-SHA-HMAC;
- an IPSec profile applied to Tunnel0.

Figures 20 and 21 present the relevant configuration and verification commands executed on R1 and R2.



```
R1#show crypto ikev2 proposal
 IKEv2 proposal: default
     Encryption : AES-CBC-256 AES-CBC-192 AES-CBC-128
     Integrity  : SHA512 SHA384 SHA256 SHA96 MD596
     PRF        : SHA512 SHA384 SHA256 SHA1 MD5
     DH Group   : DH_GROUP_1536_MODP/Group 5 DH_GROUP_1024_MODP/Group 2
 IKEv2 proposal: myProposal
     Encryption : AES-CBC-256
     Integrity  : SHA512
     PRF        : SHA512
     DH Group   : DH_GROUP_4096_MODP/Group 16
R1#
```

Figure 20: IKEv2 and IPSec configuration verification on R1



```
R2#show crypto ikev2 proposal
 IKEv2 proposal: default
     Encryption : AES-CBC-256 AES-CBC-192 AES-CBC-128
     Integrity  : SHA512 SHA384 SHA256 SHA96 MD596
     PRF        : SHA512 SHA384 SHA256 SHA1 MD5
     DH Group   : DH_GROUP_1536_MODP/Group 5 DH_GROUP_1024_MODP/Group 2
 IKEv2 proposal: myProposal
     Encryption : AES-CBC-256
     Integrity  : SHA512
     PRF        : SHA512
     DH Group   : DH_GROUP_4096_MODP/Group 16
R2#
```

Figure 21: IKEv2 and IPSec configuration verification on R2

4.3 Tunnel Establishment Verification

After configuration, successful connectivity between PC1 and PC2 was verified through the IPSec tunnel.

The outputs of the IOS verification commands confirmed:

- successful establishment of inbound and outbound IPSec Security Associations;
- packet encapsulation and decapsulation through ESP;
- operational Tunnel0 interfaces;
- successful OSPF adjacency establishment through the protected tunnel.

Figure 22 shows the active IPSec Security Associations and packet counters.

```
R1#show crypto ipsec sa

interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 1.1.1.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
   remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
   current_peer 2.2.2.2 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 41, #pkts encrypt: 41, #pkts digest: 41
    #pkts decaps: 40, #pkts decrypt: 40, #pkts verify: 40
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

     local crypto endpt.: 1.1.1.1, remote crypto endpt.: 2.2.2.2
     plaintext mtu 1438, path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/0
     current outbound spi: 0xB112069E(2970748574)
     PFS (Y/N): N, DH group: none

     inbound esp sas:
      spi: 0x5CF1C655(1559348821)
        transform: esp-aes esp-sha-hmac ,
        in use settings ={Tunnel, }
        conn id: 1, flow_id: SW:1, sibling_flags 80000040, crypto map: Tunnel0-head-0
        sa timing: remaining key lifetime (k/sec): (4333709/3289)
        IV size: 16 bytes
        replay detection support: Y
        Status: ACTIVE(ACTIVE)

     inbound ah sas:

     inbound pcp sas:

     outbound esp sas:
      spi: 0xB112069E(2970748574)
        transform: esp-aes esp-sha-hmac ,
        in use settings ={Tunnel, }
        conn id: 2, flow_id: SW:2, sibling_flags 80000040, crypto map: Tunnel0-head-0
        sa timing: remaining key lifetime (k/sec): (4333709/3289)
        IV size: 16 bytes
        replay detection support: Y
        Status: ACTIVE(ACTIVE)

     outbound ah sas:

     outbound pcp sas:
R1#
```

Figure 22: Active IPSec Security Associations

The IPSec Security Associations use different inbound and outbound SPIs, reflecting the unidirectional nature of IPSec SAs.

4.4 Analysis of IKE_SA_INIT Exchange

Traffic captures were obtained on interface G0/0 of R1 during tunnel establishment.
The IKE_SA_INIT exchange was responsible for:

- negotiation of cryptographic algorithms;
- exchange of Diffie-Hellman public values;
- generation of shared secret material;
- exchange of nonces;
- creation of the initial IKE Security Association.

Figure 23 presents the captured IKE_SA_INIT packets.

Figure 23: IKE_SA_INIT exchange captured in Wireshark

The capture shows the Security Parameter Indexes (SPIs), Security Association proposals, Diffie-Hellman key exchange payloads, and nonce payloads exchanged between both peers.
The negotiated proposal includes:

- AES-CBC-256 encryption;
- SHA512 integrity protection;
- SHA512 pseudo-random function;
- Diffie-Hellman Group 16.

The nonce payloads contribute entropy to the key generation process and ensure freshness of the exchange, protecting against replay attacks.

4.5 Analysis of IKE_AUTH Exchange

After the IKE_SA_INIT exchange, the peers performed mutual authentication using the configured pre-shared key during the IKE_AUTH exchange.
Figure 24 presents the captured IKE_AUTH packets.

`ike_auth.png`

Figure 24: IKE_AUTH exchange captured in Wireshark

Unlike the IKE_SA_INIT packets, the IKE_AUTH payloads are encrypted and authenticated. Wireshark identifies these payloads as encrypted, demonstrating that cryptographic keys derived during the previous exchange are already being used to protect the communication.

The IKE_AUTH exchange is responsible for:

- peer authentication;
- identity exchange;
- creation of the first CHILD_SA used for IPSec protection.

4.6 SA Deletion and Re-establishment

The command `clear crypto session` was executed to remove the active Security Associations and force a new negotiation process.

Wireshark captures show the deletion of the existing SAs followed by the establishment of new IKEv2 and IPSec Security Associations.

sa_reestablishment.png

Figure 25: Deletion and re-establishment of Security Associations

After the previous SAs were removed, new IKE_SA_INIT and IKE_AUTH exchanges were observed, resulting in the creation of new SPIs and new IPSec Security Associations.

The results confirm that IPSec tunnel establishment using IKEv2 was successfully achieved between R1 and R2.

The analysis of Wireshark captures demonstrated:

- negotiation of cryptographic algorithms during IKE_SA_INIT;
- exchange of Diffie-Hellman values and nonces;
- encrypted peer authentication during IKE_AUTH;
- successful creation of IPSec Security Associations;
- successful deletion and recreation of SAs after clearing the crypto session.

Cisco IOS verification commands further confirmed the operational state of the tunnel, the establishment of OSPF adjacency through Tunnel0, and the correct operation of encrypted ESP traffic.